

OMAR MOHAMED ABDELGHANI

Full-Stack Developer | Cybersecurity Background

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PROFESSIONAL SUMMARY

Computer Science student and **full-stack developer** with a **cybersecurity** background, combining hands-on Python/Java/PHP engineering with secure-systems thinking. Proven track record building **production-style backend architectures**, real-time **intrusion detection systems**, and independently designed **full-stack business platforms** (Laravel/PHP, Flask) — including a **graduation project (SpecSense AI)** led as team lead for an industrial electrical-engineering platform. Comfortable across the stack: **REST APIs, Flask, Laravel, SQL databases, Docker**, and network security tooling (**IDS/IPS, OWASP Top 10, penetration testing**). Seeking a junior full-stack or backend software engineering role where I can ship reliable, secure, well-designed systems from day one.

CORE TECHNICAL SKILLS

Languages: Python, Java, JavaScript, PHP

Backend & Web: Flask, Laravel, REST API design, HTML/CSS, session & auth handling

Databases: MySQL, SQLite, data validation & integrity pipelines

Machine Learning: scikit-learn, ML-based anomaly & threat classification

DevOps & Tools: Docker, Git/GitHub, CLI & GUI tooling

Systems & Linux: Kali Linux, shell scripting, Linux system administration, cross-platform development (Linux/Windows)

Cybersecurity: OWASP Top 10, penetration testing, Nmap, Wireshark, Burp Suite, Metasploit, IDS/IPS design, digital forensics

Networking: VLAN segmentation, OSPF routing, subnetting, Cisco Packet Tracer

EXPERIENCE

Freelance Software & AI Engineer — *Outlier*

Oct 2024 – Feb 2026

- Delivered **coding and AI-model evaluation solutions** for international clients, consistently meeting deadlines across concurrent technical engagements.
- Translated ambiguous client requirements into working technical deliverables, strengthening **independent problem-solving** and **remote collaboration** skills in a fast-paced freelance environment.

Information Technology Trainee — *Elsewedy Electric PSP*

Jul 2025 – Aug 2025

- Supported **monitoring and maintenance** of enterprise IT infrastructure across network and hardware operations for an industrial manufacturing environment.
- Assisted in **troubleshooting and system administration**, gaining direct exposure to production IT operations at an industrial-scale organization — context later applied to the SpecSense AI graduation project.

KEY PROJECTS

SpecSense AI — Graduation Project (Team Lead) | *Python, OCR, Engineering Automation* /

<https://github.com/omar230101276/SpecSense-AI>

- Architected a **modular Python backend** for an automated electrical design and calculation platform for cable sizing and electrical distribution, aligned with industrial standards used in the Elsewedy Electric context.
- Built a **high-performance calculation engine** for circuit distribution tables, voltage-drop analysis, and cable sizing estimation.
- Designed a **validation module** enforcing strict data integrity between OCR-based document extraction and the core engineering pipeline, and a **layered architecture** separating extraction, processing, and computation for scalable deployment.

EdgMon V4.0 — Mini SOC / Intrusion Detection System | *Python, Scapy, ML, Flask, SQLite* /

https://github.com/omar230101276/EdgMon_V4.0

- Engineered a **real-time hybrid IDS** combining rule-based heuristics and **scikit-learn**-based machine learning to classify network threats including **DDoS** and **port-scan attacks**.

- Built a custom **Layer-2 bidirectional flow tracker** using Scapy to extract advanced network metrics while preventing memory leaks during traffic spikes.
- Developed a **Central Threat Engine** with dynamic severity scoring, anomaly decay, and hardware-stress correlation (via psutil) to **minimize false positives**, backed by a live Flask/SQLite dashboard for alerting and trend analysis.

EdgMon V3.1 — Edge System Monitoring & Security Tool | [Python, Flask, SQLite](#) /

https://github.com/omar230101276/Automated_Server_Monitorin-Linux-Windows-

- Built a **cross-platform** (Linux and Windows) real-time edge monitoring system tracking **CPU, RAM, disk, temperature, and connectivity**, with anomaly-based alerting and structured logging for infrastructure reliability.
- Delivered a **Flask dashboard** with SQLite-backed storage to support historical analysis and audit trails.

Click-Breaker — Social Engineering Assessment Tool | [PHP, Python, Red Team](#) / [Recon](#)

- Built an **open-source security assessment tool** for social engineering and reconnaissance testing, leveraging browser-based vectors (camera, microphone, geolocation access) to simulate real-world attacks against end users.
- Designed for **authorized Red Team use**, helping organizations evaluate resilience against human-based security risks and social engineering attacks — primarily implemented in **PHP and Python**.

Multi-VLAN Enterprise Network (OSPF Routing) | [Cisco Packet Tracer, Networking](#)

- Designed and implemented a network spanning **22 VLANs, 44 endpoints, 5 routers, and 5 switches**, applying subnetting and VLAN segmentation for security and performance.
- Configured **OSPF dynamic routing**, deepening hands-on expertise in enterprise network design.

Additional Projects | [Huffman Compression](#) · [Docker E-Commerce App](#) · [Web-Based Banking Management System](#)

- Built a **Huffman-coding compression tool** in Python with entropy/compression-ratio analysis and both CLI and GUI interfaces. https://github.com/omar230101276/Huffman_Compressor
- Containerized a full **e-commerce application** (frontend, backend, database) with Docker for portable, consistent deployment. <https://github.com/omar230101276/e-commerce-microservices>
- Developed a **secure banking simulation platform** (JavaScript, Local Storage) with role-based access control, form validation, and session-style handling. https://github.com/omar230101276/Bank_Web_FrontEnd

FULL-STACK WEB DEVELOPMENT — INDEPENDENT PROJECTS

Self-initiated, end-to-end platforms designed and built from scratch (database, backend, and UI) to demonstrate independent full-stack delivery beyond coursework.

Photographer Portfolio & Booking Management System | [Laravel, PHP, MySQL](#)

- Built a full-stack platform combining a public **portfolio site**, a **client booking and inquiry portal**, and an **admin dashboard** for managing bookings, clients, and content.
- Implemented an **album/gallery system** for organizing and showcasing photo and video collections, giving photographers a client-facing way to deliver and present work.

Contracting Company Management System | [Laravel, PHP, MySQL](#)

- Developed a full-stack system covering **project/job tracking** and **invoicing**, giving a contracting business a single platform to manage active jobs and billing.
- Designed the data model and admin workflows end-to-end, from client onboarding through project completion and invoice generation.

CERTIFICATIONS & TRAINING

- TryHackMe — Jr Penetration Tester
- TryHackMe — Pre Security Path
- Elsewedy Electric PSP — Summer Internship Program 2025
- Nile University — Career Compass Workshop

EDUCATION

B.Sc. in Computer Science (In Progress)

ElSewedy University of Technology — Faculty of Engineering Technology